

Brandon McCarthy

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PROFESSIONAL SUMMARY

AI-focused Software Engineer with a background in healthcare documentation and hands-on project experience building full-stack applications, workflow automations, and LLM-assisted systems. Skilled in JavaScript, React, Node.js, Python, APIs, webhooks, n8n, and structured data workflows. Designed a human-in-the-loop lead intake automation system that uses LLM-based classification, CRM-style tracking, and AI-generated response drafts to reduce repetitive manual review. Seeking to contribute to healthcare AI products that improve operational efficiency, data quality, and user workflows.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, SQL
- **AI / ML:** Scikit-learn, Pandas, NumPy, Data Analysis, Data Visualization, Prompt Engineering, Retrieval-Augmented Generation (RAG), LLM Applications
- **Databases:** PostgreSQL, SQLite
- **Frameworks / Libraries:** React, Redux, Express.js, Flask
- **Tools:** Git, GitHub, Docker, Postman, VS Code, Jupyter Notebook
- **Cloud / APIs:** AWS, REST APIs, API Integration
- **Domain / Core Skills:** Healthcare Data, Data Pipelines, Workflow Automation, Clinical Documentation, Data Quality, Debugging

PROJECT EXPERIENCE

McCarthy AI Automations <https://www.mccarthyaiautomations.com/>

- Built and deployed AI-powered automation systems using LLMs (OpenAI, Claude) to process, classify, and act on structured and unstructured data.
- Designed and implemented retrieval-augmented generation (RAG) workflows to support context-aware responses using project, support, and service data.
- Developed end-to-end pipelines integrating APIs, databases, and AI models to automate lead intake, classification, follow-up, and internal workflow management.
- Created scalable client-facing dashboards and internal admin tools for tracking projects, milestones, billing, and support requests.

DineAir <https://www.dineair.onrender.com>

- Built a full-stack, real-time delivery platform supporting multiple user roles, designing scalable data models and role-based access control for complex workflows.
- Engineered real-time data pipelines using WebSockets and integrated external APIs including Google Maps and flight data to process live location and status updates.
- Designed and optimized backend services and database interactions to handle high-frequency transactional data, ensuring data integrity, validation, and system reliability.

StayHaven <https://www.stayhaven-hyyh.onrender.com>

- Built a full-stack application with React, Express, and PostgreSQL, supporting structured data workflows and user-driven interactions.
- Designed RESTful APIs and optimized database queries to manage and retrieve application data efficiently.
- Implemented authentication, validation, and data integrity checks to ensure secure and reliable data handling.

PROFESSIONAL EXPERIENCE

Physician Life Care Planning

Sep 2023 - Present

Production Specialist / Transcriptionist

Remote

- Analyzed and structured complex medical records into clear, actionable reports for legal and healthcare stakeholders.
- Processed high volumes of unstructured clinical documentation, transforming information into organized, searchable formats.
- Collaborated with physicians to extract key findings from clinical data in support of evidence-based case development.

AQuity Solutions

Jul 2021 - Sep 2023

Medical Scribe

Remote

- Managed and validated structured and unstructured clinical data, ensuring accuracy, consistency, and compliance with healthcare standards.
- Identified documentation inconsistencies and improved data quality through workflow optimization.
- Supported high-volume medical record review and data entry in the EMR system, maintaining data integrity and enabling clinicians to access up-to-date charts promptly

EDUCATION

Coursera

Jul 2025 - Aug 2025

Machine Learning Specialization, Machine Learning

App Academy

Mar 2024 - May 2025

Full Stack Web Development Bootcamp, Full Stack Web Development

Hack Reactor

Jun 2023 - Oct 2023

Full Stack Web Development Bootcamp, Full Stack Web Development

East Tennessee State University

Aug 2014 - May 2018

Bachelor of Science, Biology, Psychology